

Assignment – 4.2

Q - 1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. No display	Blank screen, no display
2. RAM failure	PC does not boot or beeps
3. Hard disk failure	Slow boot or “disk not found”
4. Overheating	PC shuts down/restarts
5. Power supply failure	PC does not turn on

Q - 2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Laptop Not Charging – Troubleshooting Steps:

1. **Check power outlet** – Make sure the socket is working.
2. **Check charger** – Look for damaged cables, plug, or adapter.
3. **Check charging port** – Remove dust and check for loose or bent pins.
4. **Reconnect charger** – Ensure the charger is firmly connected.
5. **Try another compatible charger** – Check if the laptop starts charging.
6. **Test with multimeter** – Check whether the adapter gives the voltage shown on its label.
7. **Check battery** – Look for swelling or physical damage.
8. **Power reset** – Disconnect power, hold the power button for 15–30 seconds, then reconnect.
9. **Check charging jack** – If it is loose or damaged, it may need repair.

Conclusion: If still not charging, the battery or internal charging circuit may be faulty.

Q - 3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

USB Charger Voltage Test:

- Measured voltage: **5.02 V**
- Expected range: **4.75–5.25 V**
- **Result:** Within the expected range — charger voltage is normal.

Q - 4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Testing the PSU 12V Rail:

1. Disconnect the ATX power connector from the motherboard.
2. Set the multimeter to **DC Voltage (V --)**.
3. Connect the **black probe to a black (GND) wire**.
4. Connect the **red probe to the yellow (+12V) wire**.
5. Turn on the PSU/PC as appropriate for the test and read the voltage.

Measured voltage: 12.1 V

Expected range: About 11.5–12.6 V.

If the voltage is below 11.5 V: It may indicate a **weak or faulty power supply (PSU)**, which can cause random shutdowns.